



NNP Pocket Guide 2

FEN, CV, Resp, and Rx

Version 6, October 2026

Companion to the procedures booklet; same ring.

Sections

-  FEN: GIR, TPN, calories, bilirubin
-  CV: pressors, profiles, BP norms
-  Resp: blood gas, vent
-  Rx: antibiotics, levels, pain, drips, emergency drugs (Baylor)
-  Ref: sources

Orange  pages follow their topic and show where Baylor/TCH practice differs.

Starting points, not orders. Follow unit protocol and your attending.



GIR (mg/kg/min)

Shortest	$\%D \times \text{mL/kg/day} \div 144$
Hourly	$\%D \times \text{mL/hr} \div (6 \times \text{kg})$ $= \%D \times \text{mL/hr} \times 0.167 \div \text{kg}$
Long way	$\text{mL/day} \times \%D \text{ (decimal)} = \text{g}; \times 1000 = \text{mg}; \div \text{kg} \div 1440$

%D for a target GIR = $\text{GIR} \times 144 \div \text{mL/kg/day}$

Example: D10 at 80 mL/kg/day = GIR 5.6.

Targets

Start 4–6, then titrate to glucose; preterm infants need at least 4.

Max dextrose

12.5% peripheral. 25% central with the tip in the SVC.



Glucose

Baylor

FEN

First IV fluid	D10 at 65–80 mL/kg/day (GIR $\approx$ 4.5–5.5)
NPO newborn	GIR 5–8; glucose by 30–60 min (<25 wk: 4–6)
Starting PN	GIR 4–6 (some ELBW only 3.5–4.5 at first)
Goal on full PN	GIR 10–12
Max dextrose	12.5% peripheral; up to 30% central

Low and can't feed: D10 2 mL/kg, then GIR up 10–20%. **Persistent** (past 48–72 h): goal $\geq$ 60, or $\geq$ 70 if a genetic cause. Endocrine if GIR >10 and can't wean.

High (>150 persistently): GIR down 1–2, not below 3.5; no dextrose under 5%. **Insulin** if >180 on a low GIR (optional bolus 0.05–0.1 units/kg over 15 min, then infusion); keep under 180.



TPN: start and advance

When: <32 wk, VLBW/ELBW, GI anomaly or surgery, NPO >3–5 days.

Day 1: D10 at 60–80 mL/kg/d, AA 2–3 g/kg/d, lipid 1 g/kg/d.

Amino acids	Within 24 h at 2–3; +0.5/d to 3–4, max 4
Lipid	1–2 → 3 g/kg/d. 0.5–1 prevents EFAD. TG >200: stop or cut 0.5–1.
Sodium	None first 24 h, then 2–4 mEq/kg/d
Potassium	None first 24–48 h, then 1–3 mEq/kg/d once urine output is established
Ca gluconate	200–400 mg/kg/d

Peripheral TPN under 900 mOsm/L.



PN start and goals Baylor

	Start	Goal
Protein	2–3 g/kg	3.5–4 preterm; 2–3 term
Lipid	by weight, below	3 g/kg
GIR	4–6	10–12
Ca and Phos	1–1.2 mmol/kg	1.5–2
Na and K	0	2–4 mEq/kg
PN energy	42–57 kcal/kg	90–100

Lipid start: ≤ 750 g 0.5–1 g/kg; 751–1000 g (or SGA, IUGR, steroids, sepsis) 1 g/kg; AGA >1000 g 2 g/kg. Advance 1 g/kg/day only if TG <250 . Cysteine 30–40 mg/g amino acids. PN at 130 mL/kg/day meets most needs.



TPN calories

Glucose	3.4 kcal/g
Protein	4 kcal/g
Fat	9 kcal/g
20% lipid	1 g = 5 mL = 10 kcal (2 kcal/mL)

$$\text{Total} = \text{GIR} \times 5 + \text{AA g/kg} \times 4 + \text{lipid g/kg} \times 10$$

Exact dextrose kcal/kg/day = %D $\times$ mL/kg/day $\times$ 0.034 (GIR $\times$ 4.9).

$$\text{Lipid g/kg/day} = \text{mL/hr} \times 24 \times 0.2 \div \text{kg}$$

Example: 1.8 kg, D12 at 10.2 mL/hr, AA 3.5, lipid 1.1 mL/hr $\approx$ 99 kcal/kg/day.

Goals

Preterm: 115–140 kcal/kg/day on full feeds; about 90–120 on TPN alone. Term: 100–120. Gain 15–20 g/kg/day.

$$\text{Feeds: kcal/mL} = \text{kcal/oz} \div 30$$

20 $\rightarrow$ 0.67	22 $\rightarrow$ 0.73	24 $\rightarrow$ 0.8
26 $\rightarrow$ 0.87	28 $\rightarrow$ 0.93	30 $\rightarrow$ 1.0



Feeds and vitamins Baylor

Birth weight	Start	Advance
≤1250 g	20 trophic (3–5 days)	20/day
1251–1500 g	20	30/day
1501–2000 g	20	30–40/day
2001–2500 g	25–30	30–40/day

mL/kg/day. Start within 6–12 h if stable. Fortify after 1 day at 80 mL/kg/day; Prolacta for ≤1250 g. Full feeds ~160 fortified human milk or 150 preterm formula.

Energy 110–130 kcal/kg/day enteral; PN goal 90–100. Growth 15–20 g/kg/day (<2 kg).

Iron 2–3 mg/kg/day on fortified milk, from day 14. **Vitamin D** 200 IU/day on some fortifiers; term breastfed 400.



Bilirubin

35 weeks and up

Thresholds are hour-specific curves by GA and risk: plot them in BiliTool or PediTools.

Neurotoxicity risk factors: GA <38 wk, albumin <3.0, hemolysis (isoimmune, G6PD), sepsis, instability in the past 24 h.

Escalation: TSB within 2 of the exchange line = emergency. NICU, intensive phototherapy, PO + IV fluids, STAT labs, TSB every 2 h.

Stop phototherapy at ≥ 2 below the start threshold; rebound in ~24 h (6–12 h if started before 48 h).

Under 35 weeks (Maisels 2012)

PMA	Photo	Exchange
<28 wk	5–6	11–14
28–29	6–8	12–14
30–31	8–10	13–16
32–33	10–12	15–18
34–35	12–14	17–19

TSB mg/dL. Born <35 wk: stay on the top row after 35 wk PMA.



Bilirubin Baylor

Under 35 weeks, by birth weight (TSB mg/dL):

BW	Phototherapy	Exchange
<750 g	≥ 5	>13
750–999	≥ 5 ; wk 2 ≥ 7	>15
1000–1499	5–7; to 1 wk 7–9; wk 2 10–12	15–16
1500–1999	5–8; 8–10; 10–12; wk 2 13–15	16–18
2000–2500	5–8; 8–11; 11–13; 13–15; wk 2 14–15	18–19

Phototherapy steps: <24 h, 24–48, 48–72, 72–168 h, week 2. Lower for sick infants.

IVIG (isoimmune, rising despite intensive phototherapy or within 2–3 of exchange): 0.5–1 g/kg over 2 h; repeat at 12 h.

B/A ratio for exchange: 8.0 (≥ 38 wk, no risk factors); 7.2; 6.8 (35–37 wk with risk factors).



Pick the pressor

BP = CO × SVR: pipes, pump, or both?

Low diastolic	Pipes (low SVR) → vasopressor ($\alpha 1$ or V1)
Low systolic	Pump (low contractility) → inotrope ($\beta 1$ or PDE3)
Both	Cold, mottled, acidotic → both actions

	Warm (low SVR)	Cold (high SVR)
Skin	Warm, pink	Cool, mottled
Refill	Flash <2 s	>3 s
Pulses	Bounding	Thready
Pulse pressure	Wide	Narrow
Causes	Sepsis, NEC, big PDA	Cold sepsis, transitional, post-ligation

Hypotensive, warm, and bounding is **not** reassuring: it's vasodilated and needs tone.



Pressor doses

Drug	Dose	Use
Dopamine	≤ 2 / ≥ 5 –10 / ≥ 10 mcg/kg/min	DA $\rightarrow$ β_1 $\rightarrow$ α_1 with dose. ≥ 10 : R $\rightarrow$ L ductal shunt risk. Not with PPHN.
Dobutamine	≥ 3 –15 mcg/kg/min	β_1 inotrope, mild β_2 dilation. Avoid in outflow obstruction (IDM).
Epinephrine	0.02–0.1, then ≥ 0.1 (caution >0.2)	Low dose β , high dose α_1 . Lactate, glucose, tachycardia.
Norepinephrine	0.02–0.4 mcg/kg/min	$\alpha_1 > \beta_1$. Severe vasoplegia.
Vasopressin	0.0002–0.005 units/kg/min	V1a. Catecholamine-resistant; spares PVR. Low Na, low platelets.
Milrinone	0.25–0.75 mcg/kg/min, no load	PDE3 inodilator. Low output + PH, post-ligation. Onset 1–2 h; drops BP.
Hydrocortisone	1 mg/kg IV q8h	Adrenal insufficiency, or refractory on ≥ 2 agents.



Pressors Baylor

Drug	Start	Range
Dopamine	5 mcg/kg/min	Add an agent if no effect at 10–15
Dobutamine	5	2–15
Epinephrine	0.01–0.03	0.01–0.3
Milrinone	0.375, no load	0.375–0.75; BP must be adequate; not with oliguria
Vasopressin	0.01 units/kg/hr	0.005–0.04 units/kg/hr

Hydrocortisone: 1 mg/kg IV once; if hypotension persists or recurs over 4–6 h, 1 mg/kg every 8 h for 24–48 h, off by day 5. No routine cortisol levels.

No volume boluses without real hypovolemia.



Decision tree and algorithm

1. **Bed loose** (warm, low diastolic) → norepinephrine (or dopamine)
2. **Pump weak** (cool, poor contractility) → dobutamine
3. **Pump weak + high afterload** (PPHN, post-op) → milrinone, add a pressor if BP sags
4. **Both failing** → epinephrine
5. **Still low on catecholamines** → vasopressin; think hydrocortisone

Around it

Perfusion adequate (NIRS, urine output, lactate, echo)? Then permissive hypotension. Reassess after every change; BP up with worse perfusion means over-constriction. Wean catecholamines first, keep milrinone while PH persists, wean vasopressin last.



Hemodynamic profiles

Read diastolic BP, pulse pressure, perfusion index, SVR, cardiac output, and StO₂ together.

Profile	Pattern
Vasodilatory	↓ DBP, ↓ SVR; ↑ PP, PI, CO (warm shock)
Vasoconstrictor	↑ DBP, ↑ SVR; ↓ PP, PI, CO (cold shock)
Cardiogenic	↓ CO, low PP, PI, DBP
L-R shunt	↑ CO, ↑ PP; ↓ DBP, ↓ SVR (PDA run-off)
Volume depletion	↓ CO, ↓ PP; ↑ SVR

Same low diastolic, opposite fixes: vasodilatory needs a vasoconstrictor; volume-depleted needs volume.

Epinephrine 0.2 mcg/kg/min for an hour $\approx$ one resuscitation bolus (10 mcg/kg).



BP norms (after 2 weeks)

By PMA; 50th / 95th / 99th percentile, mmHg (Dionne 2012, corrected table).

PMA	MAP	Systolic	Diastolic
44	63/80/85	88/105/110	50/68/73
42	62/76/81	85/98/102	50/65/70
40	60/75/80	80/95/100	50/65/70
38	59/74/79	77/92/97	50/65/70
36	57/72/77	72/87/92	50/65/70
34	50/65/70	70/85/90	40/55/60
32	49/64/69	68/83/88	40/55/60
30	48/63/68	65/80/85	40/55/60
28	45/58/63	60/75/80	38/50/54
26	38/57/63	55/72/77	30/50/56

High BP: $\geq$ 95th on 3 occasions; $>$ 99th, investigate. **"MAP $\geq$ GA":** weak evidence; judge with perfusion. **Cuff:** compare MAP (the measured value), right arm.



Read a blood gas

1. **pH:** <7.35 acidemic, >7.45 alkalemic
2. **PaCO_2 :** respiratory?
3. **HCO_3^- / BE:** metabolic?
4. **Compensation, then the bedside picture**

CO_2 opposite pH $\rightarrow$ respiratory. HCO_3^- with pH $\rightarrow$ metabolic.

pH	7.35–7.45 (>7.30 may be OK in context)
PaCO_2	35–45
HCO_3^-	22–26
BE	-2 to +2; worse than -4 to -6, watch

Uncompensated	pH off, other value normal
Partial	pH off, both abnormal
Compensated	pH normal, both abnormal

Lungs fix CO_2 quickly; kidneys fix HCO_3^- over days. High CO_2 **and** low HCO_3^- = mixed acidosis, not compensation.



What moves what

CO₂ ↑	pH ↓ (respiratory acidosis)
CO₂ ↓	pH ↑ (respiratory alkalosis)
HCO₃⁻ ↓ (lactate, TPN acid, RTA, diarrhea)	pH ↓, BE more negative
HCO₃⁻ ↑ (vomiting, diuretics, bicarb, acetate)	pH ↑
Bicarb given	HCO ₃ ⁻ + H ⁺ → carbonic acid → CO₂ + water : CO ₂ rises; pH can fall if not ventilated
CO₂ high for days	Kidneys keep HCO ₃ ⁻ → pH toward normal
Metabolic acidosis	Lungs blow off CO ₂ quickly
Rate or tidal volume ↑	CO ₂ ↓, pH ↑
MAP or FiO₂ ↑	Oxygenation ↑

Bicarb: only with adequate ventilation; not in the same line as catecholamines; can lower K.



Gases and modes Baylor

$$\Delta pH = \Delta PCO_2 \times 0.008$$

PCO_2 about 60 is acceptable; correct $PCO_2 < 35$ promptly.

Default mode ≤ 32 wk: pressure-control A/C with volume guarantee.

NIPPV	Primary	Post-extubation
PIP	20–22	2 above vent PIP
PEEP	5–6	Same as vent
Rate / Ti	20 / 0.5	20 / 0.5



Vent modes

AC	Every breath supported. Poor effort, apnea, extreme prematurity.
SIMV	Synced breaths plus spontaneous. Weaning, low support, sedation backup.
Volume guarantee	Hits a set volume; unstable compliance; guards against volutrauma.
Pressure control	Caps pressure; volume varies. RDS, severe BPD.
HFOV	Open lung, tiny volumes. PIE, air leak, PPHN.
HFJV	Fast small bursts; conventional vent gives PEEP. PIE, refractory CO ₂ .

By disease

RDS: CPAP → PC or volume-targeted; HFOV if severe. **TTN:** HFNC or CPAP. **BPD:** long expiratory time, permissive CO₂, often VG. **PIE/air leak:** HFJV or HFOV. **Pulmonary hemorrhage:** higher MAP. **PPHN:** conventional or HFOV.



What to change: conventional and HFOV

Conventional

CO₂ high	Rate +5–10; PIP +2 for chest rise; lower PEEP if over-expanded
O₂ low	FiO ₂ ↑; PEEP ↑; PIP +1–2; Ti 0.4–0.6 s
Weaning	FiO ₂ to 80% before changing PIP, I:E, PEEP; steps PIP 2, FiO ₂ 5%, rate 5

$$\text{MAP} = (\text{PIP} - \text{PEEP}) \times \text{Ti} \div (\text{Ti} + \text{Te}) + \text{PEEP}$$

HFOV

CO₂ high	Amplitude ↑; lower Hz if needed
CO₂ low	Amplitude ↓; raise Hz
O₂ low	MAP ↑; FiO ₂ ↑
O₂ high	FiO ₂ ↓, then MAP ↓

Lower Hz = bigger volumes, more CO₂ removal; Hz changes are usually a senior call.



Volume guarantee Baylor

Target Vt 4.5–5.0 mL/kg <1000 g; 4.0–4.5 >1000 g; change by 0.5. Pmax 25–28, then 3–5 above working PIP. PEEP $\geq$ 5. Ti 0.3. Backup rate 30. Flow 6–8 L/min.

Weaning: wean FiO₂ and target Vt; don't go to $\leq$ 4 mL/kg; consider extubation instead.

Ready to extubate

VG modes	Vt 4–5 mL/kg with PIP <25
PC modes	PIP 20
All	MAP 8–10, PEEP 5–6, FiO ₂ $\leq$ 30%; breathing above backup 30 (A/C) or rate 20–25 (SIMV)



Jet, weaning, and OI

HFJV

Ventilation: PIP above PEEP (ΔP). Oxygenation: PEEP (+1–2 when $FiO_2 \geq 0.6$). Don't wean PEEP until $FiO_2 < 0.35$ –0.40.

HFOV weaning

- $PaCO_2 < 50$ and $pH > 7.25$: wean amplitude
- $FiO_2 < 70\%$, $PaO_2 > 60$, good inflation: wean MAP
- Extubation: term MAP 11–14, amp ≤ 20 ; LBW MAP 10–12, ≤ 15 ; VLBW MAP < 10 , < 12
- Air leak: wean MAP hard, smallest amplitude, accept CO_2 high 50s–low 60s, PIE side down

$$OI = MAP \times FiO_2 \times 100 \div PaO_2$$

OI > 40 for > 2 h, or in 3 of 5 gases: associated with about 80% mortality.



Jet and resp meds Baylor

Jet start: rate 360–420, Ti 0.02, PIP 20–25, PEEP 5–12 (from conventional MAP). Conventional breaths rate 0–5, Ti 0.4–0.6, PIP half the jet PIP or 5–8 above PEEP. Back to conventional when air leak resolved, $\Delta P < 30$, gases stable.

Curosurf 2.5 mL/kg, then up to 2 doses of 1.25 mL/kg 12 h apart.

Caffeine routinely for all < 1250 g.

iNO 20 ppm after lung recruitment; response = $\text{PaO}_2 + 10$ or $\text{SpO}_2 + 5\%$ at 30 min. Wean carefully below 5 ppm (1 $\rightarrow$ 0.5 before stopping). FiO_2 may need +15% after.



Ampicillin, penicillin, nafcillin Baylor

GA at birth; PNA = days old.

Ampicillin EOS	100 mg/kg every 8 h
Amp meningitis (incl. GBS)	PNA ≤ 7 d: 100 mg/kg q8h; >7 d: 75 mg/kg q6h
Amp other, ≤ 34 wk	50 mg/kg q12h (≤ 7 d); 75 mg/kg q12h (>7 d)
Amp other, 35–43 wk	50 mg/kg q8h
UTI prophylaxis	Amp 25 mg/kg IV q24h, or amoxicillin 10 mg/kg PO daily
PCN G, GBS (not CNS)	50,000 units/kg: q12h (≤ 7 d), q8h (8–28 d), q6h (29–60 d)
PCN G, GBS meningitis	0–7 d: 84,000–150,000 units/kg q8h; 8–28 d: 100,000–125,000 q6h; 29–60 d: 100,000 q6h
PCN G, other meningitis	50,000 units/kg: q8h (0–7 d), q6h (8–28 d)
Nafcillin ≤ 34 wk	25 mg/kg q12h (≤ 7 d); q8h (>7 d)
Nafcillin 35–43 wk	25 mg/kg q8h (≤ 7 d); q6h (>7 d)
Nafcillin meningitis	25 mg/kg q8h (≤ 7 d); 25–37.5 mg/kg q6h (>7 d)



Gentamicin, amikacin, vancomycin

Baylor

Gent <30 wk	5 mg/kg q48h (PNA $\leq$ 14 d); q36h (>14 d)
Gent 30–34 wk	5 mg/kg q36h ($\leq$ 14 d); q24h (>14 d)
Gent 35–43 wk	4 mg/kg q24h ($\leq$ 7 d); 5 mg/kg q24h (>7 d)
Amikacin <30 wk	15 mg/kg q48h ($\leq$ 14 d); q24h (>14 d)
Amikacin 30–34 wk	15 mg/kg q24h ($\leq$ 60 d)
Amikacin 35–43 wk	15 mg/kg q24h ($\leq$ 7 d); 17.5 mg/kg q24h (>7 d)
Vanc <1.2 kg	15 mg/kg q18–24h (PNA $\leq$ 28 d)
Vanc 1.2–2 kg	15 mg/kg q12–18h (<7 d); q8–12h ($\geq$ 7 d)
Vanc >2 kg	15 mg/kg q8–12h (<7 d); q6–8h ($\geq$ 7 d)

Aminoglycoside levels if ELBW, renal dysfunction, or treating >72 h.



Ceftazidime, clindamycin, infiltration

Baylor

Ceftaz <1 kg	50 mg/kg q12h (≤ 14 d); q8h (>14 d)
Ceftaz 1–2 kg	50 mg/kg q12h (≤ 7 d); q8–12h (>7 d)
Ceftaz >2 kg	50 mg/kg q12h (≤ 7 d); q8h (8–60 d)
Ceftaz meningitis	50 mg/kg q8–12h (≤ 7 d); q8h (>7 d)
Clinda <1 kg	5 mg/kg q12h (≤ 14 d); q8h (>14 d)
Clinda 1–2 kg	5 mg/kg q12h (≤ 7 d); q8h (>7 d)
Clinda >2 kg	5 mg/kg q8h (≤ 7 d)

IV infiltration

Stop and disconnect; don't flush. Elevate. No heat, especially moist heat. **Hyaluronidase** (hypertonic fluids only, not pressors): five 0.2 mL injections of 15 units/mL around the leading edge, best within 1 h.



Drug levels Baylor

Drug	Goal (mcg/mL)
Gentamicin	Peak 8–10 (life-threatening), 6–10 (serious), 4–6 (UTI). Trough <1.5
Amikacin	Peak 25–40, 20–25, 15–20. Trough <8
Vancomycin	Trough 8–20 by indication and site
Phenobarbital	20–40
Phenytoin	Total 15–20; free 1–2 (toxic >4); order both

Peak: 30 min after the infusion ends. Trough: 30 min before the next dose. Peaks show efficacy; troughs show safety. Hold a dose for the trough result only if drawn for renal dysfunction.



Pain and sedation Baylor

Morphine IV	0.05–0.1 mg/kg over 5–10 min q4–6h
Morphine PO	0.08–0.1 mg/kg q4–6h
Morphine drip	Load 0.05–0.1 mg/kg, then 0.01–0.02 mg/kg/hr; up $\leq$ 0.02 q30 min
Fentanyl IV	1–2 mcg/kg over 5 min q2–4h
Fentanyl drip	0.5–1 mcg/kg/hr; up $\leq$ 1 q30 min
Lorazepam	0.05 mg/kg (0.02–0.1) q4–8h; status 0.1 mg/kg
Midazolam	0.05–0.15 mg/kg q2–4h; drip 0.03 ($\leq$ 32 wk PMA) or 0.06 mg/kg/hr

24% sucrose 2 min before: <35 wk 0.2 mL q2 min $\times$ 3 (max 0.6 mL); ≥ 35 wk 1 mL $\times$ 3 (max 3 mL); 3 series/24 h.

Fentanyl preferred in renal dysfunction; bolus <5 min can cause rigidity.
Avoid benzodiazepines <44 wk PMA per the chart.



Emergency drugs Baylor

Adenosine	0.1 mg/kg, then 0.2; rapid push, flush before and after
Albumin 5%	10–20 mL/kg over 2–4 h (volume only)
CaCl 10%	20 mg/kg push over 3–5 min
Ca gluconate	100 mg/kg over 5–10 min
Cardioversion	0.5–1 J/kg, then 2 J/kg (synchronized)
D10	Low glucose 2 mL/kg; high K 4 mL/kg + insulin 0.1 units/kg
Epi 0.1 mg/mL	IV 0.1–0.3 mL/kg; ET 0.5–1 mL/kg; max 1 mL
Lidocaine	IV 1 mg/kg; ET 2 mg/kg (VT/VF, not SVT)
Naloxone	0.1 mg/kg IV/IM q2–3 min; ET 0.2
Bicarb 4.2%	2 mEq/kg over 2 min; discouraged in codes

Calcium never with phosphate fluids. ET doses: 5 manual breaths after.



Infusions Baylor

Alprostadil	0.0125–0.1 mcg/kg/min
Dopamine	2.5–20 mcg/kg/min
Epinephrine	0.01–1 mcg/kg/min
Norepinephrine	0.02–1 mcg/kg/min
Phenylephrine	0.1–0.5 mcg/kg/min
Vasopressin	0.01–0.04 units/kg/hr
Milrinone	0.25–0.75 mcg/kg/min
Nitroprusside	0.5–10 mcg/kg/min (<3 in renal dysfunction)
Nicardipine	0.5–2 mcg/kg/min
Esmolol	25–500 mcg/kg/min

$$\text{mL/hr} = \text{dose} \times \text{kg} \times 60 \div \text{concentration}$$

For per-minute doses with concentration in the same unit (mcg). Chart ranges; usual pressor starts are on the CV pages.



Intubation and other drugs Baylor

Atropine	0.02 mg/kg rapid push; no minimum
Fentanyl 5 mcg/mL	1–2 mcg/kg over 5 min if no paralytic; intranasal 1.5–2 mcg/kg
Vecuronium	0.08–0.1 mg/kg rapid push; ongoing 0.1 q1–2h PRN; drip 0.06–0.09 mg/kg/hr
Ibuprofen lysine (PDA)	10 mg/kg, then 5 mg/kg q24h × 2 (birth weight)
Indomethacin (IVH prophylaxis)	0.1 mg/kg q24h × 3 within 12 h; ≤26 6/7 wk or <800 g

Fentanyl 5 mcg/mL = 1 mL of 50 mcg/mL + 9 mL NS. Vecuronium 1 mg/mL = 10 mL sterile water in a 10 mg vial.



Sources

Course lectures and handouts: Intro to Neonatal FEN; Neonatal TPN; FEN formulas; Chorioamnionitis and EOS screening; Cardiovascular Function and Treatments; Choosing a Pressor; Neonatal Hemodynamics; Module 2 Respiratory; Blood Gas Interpretation. Program NNP Brain reference; NNP course quick-reference card.

Baylor and Rx pages: Baylor College of Medicine, Guidelines for Acute Care of the Neonate, Ed. 34 (2026–2027), Ch. 3, 4, 8, 9, 12, 14, 16, 19.

Nutrition: ESPGHAN 2022 enteral nutrition position paper; Clinical Nutrition ESPEN, Principles of feeding the preterm infant.

Bilirubin: AAP 2022 Clinical Practice Guideline (Kemper et al.); Maisels et al., J Perinatol 2012; UNC NCCC hyperbilirubinemia guideline (2025).

BP: Dionne, Abitbol, Flynn 2012 (erratum) via Batton, Clin Perinatol 2020; Starship and Medscape neonatal hypertension.

Vent: Loyola neonatal ventilation supplement; SPA oscillator review; UNC NCCC high-frequency ventilation; UW Madison NICU jet guideline; NHS GGC HFOV.

Bicarb: Canterbury DHB and NSW ANMF neonatal guidelines.

Full versions, calculators, and links are in the NNP app.